

Observed Reddit posting around two coding-tool preview announcements

Nickolas K. Freeman

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Abstract

We examine daily posting counts associated with pseudonymous identifiers supplied as Reddit accounts. The file contains 283,488,780 identifier-day records and 479,582,977 counted posts from 2006-04-23 through 2026-09-22. We describe posting volume and concentration, then estimate local changes around the February 2025 Claude Code and May 2025 cloud Codex research-preview announcements. The 30-day estimates are 0.16% and -0.62%, respectively, with confidence intervals that include zero. Estimates change with the chosen window and trend. Unknown collection coverage and the absence of content or a comparison series prevent attribution to coding tools or identification of automated accounts.

1 Research question

Agentic coding tools can create and publish text through multistep workflows. This capacity raises a question for researchers who use platform records: did posting patterns change as these tools became available? Counts alone offer a useful first description, but they cannot identify how a post was written. The outcome can also change when a platform’s users, policies, or data collection change.

Prior research examines distinct outcomes that count data can conflate. Burtch, Lee, and Chen found a decline in Stack Overflow activity after ChatGPT’s release but did not detect the same decline in the Reddit developer communities they examined [1]. Sun and colleagues used text-based detection to estimate a slower rise in machine-generated text on Reddit than on Medium and Quora during 2022–2024 [2]. Their detector and sample differ from the identifier-day counts used here. These studies motivate an examination of posting volume without making volume a measure of machine-generated text.

2 Data and methods

The supplied Parquet file has three fields: an integer identifier, a Coordinated Universal Time (UTC) calendar date, and a positive daily count labeled posts. The supplier describes identifiers as consistent for an account across dates, but the file cannot validate that mapping or link identifiers to named accounts. The file does not define whether counted items include comments. It also lacks communities, account metadata, sampling rules, collection logs, and a denominator of all Reddit posts. We treat its values as the observed corpus, not as a verified census of the platform. The source file remains unchanged; analysis code reads it in Parquet row groups and writes separate derived summaries.

We checked the account-date sort order, duplicate account-date keys, nulls, and nonpositive counts. We aggregated posts and distinct active accounts by date. We also calculated total posts, active days, and the maximum daily count for each account. A high-volume day is defined as at least 20 posts from one account. This threshold describes the upper tail and tests sensitivity to prolific accounts. It does not classify an account as a bot.

The raw file spans 2006-04-23 to 2026-09-22. It contains 6,724 observed dates among 7,458 calendar days. The early years have dates with no observed rows, which we do not recode as zero posting. The 2025 onward segment has 627 consecutive observed dates through 2026-09-19. We exclude September 20–22, 2026 from trend exhibits because the counts fall rapidly at the file boundary: 688,357 posts on September 19 and 9,470 on September 22. This exclusion does not affect the 2025 release-window models.

Anthropic announced a limited Claude Code research preview on February 24, 2025 [3]. OpenAI announced a cloud Codex research preview on May 16, 2025 [4]. OpenAI had already introduced Codex’s command-line interface (CLI) in April [5], so May 16 is not the first availability of a Codex tool. Neither date measures Reddit adoption. For each announcement, we fit a separate ordinary least squares model to 30 preceding and 30 following complete days, omitting the announcement day:

$$\log(1 + P_t) = \alpha + \beta_1 t + \beta_2 I(t > 0) + \beta_3 \max(t, 0) + \sum_{d=1}^6 \gamma_d D_{dt} + \epsilon_t.$$

Here, P_t is the observed daily post total, t is days relative to the announcement, $I(t > 0)$ marks the following period, and D_{dt} are day-of-week indicators. We interpret $100(\exp(\beta_2) - 1)$ as the modeled immediate percentage change. Confidence intervals use heteroskedasticity and autocorrelation consistent standard errors with seven lags. We adjust the two primary p values with Holm’s method. The planned 30-day window focuses on the local period; we also fit 14-, 21-, 45-, and 60-day windows, a separate quadratic trend in the 30-day window, and a fit excluding identifier-days with at least 20 posts. These variants probe dependence on window, trend, and prolific days. The model describes a local discontinuity under its assumptions; it does not estimate a causal effect of either product.

3 Results

The full file contains 47,041,284 distinct identifiers and 479,582,977 posts. The checks found no null cells, nonpositive counts, duplicate account-date rows, or violations of account-date order. The earliest years are sparse, so Figure 1 shows observed monthly totals without filling absent dates. The corpus grows sharply during late 2024. Collection rules are unknown, so this rise cannot be interpreted as platform-wide growth. The plotted series ends with the last complete calendar month.

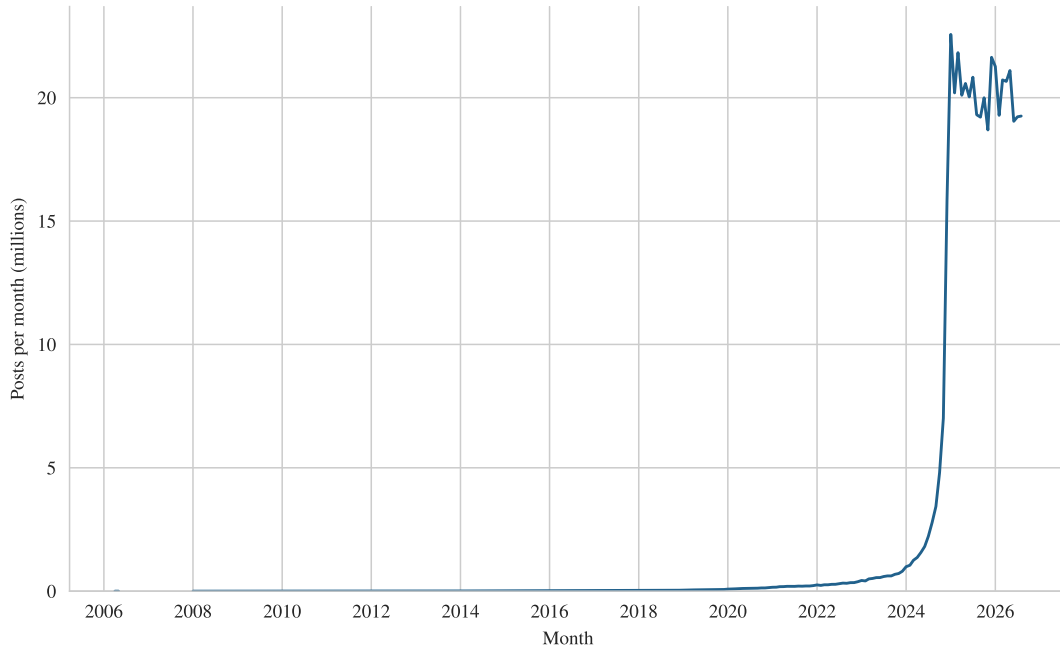


Figure 1: Observed monthly posts in the supplied file through August 2026. Months without rows remain gaps.

Figure 2 shows seven-day trailing means of daily posts and active accounts from January 2025 through September 19, 2026. The dashed and dotted lines identify release announcements, not changes inferred from the series. Mean daily volume was 671,161 posts in 2025 and 662,216 over the observed portion of 2026. The two series move together, which makes it useful to inspect both posting frequency and the number of active accounts.

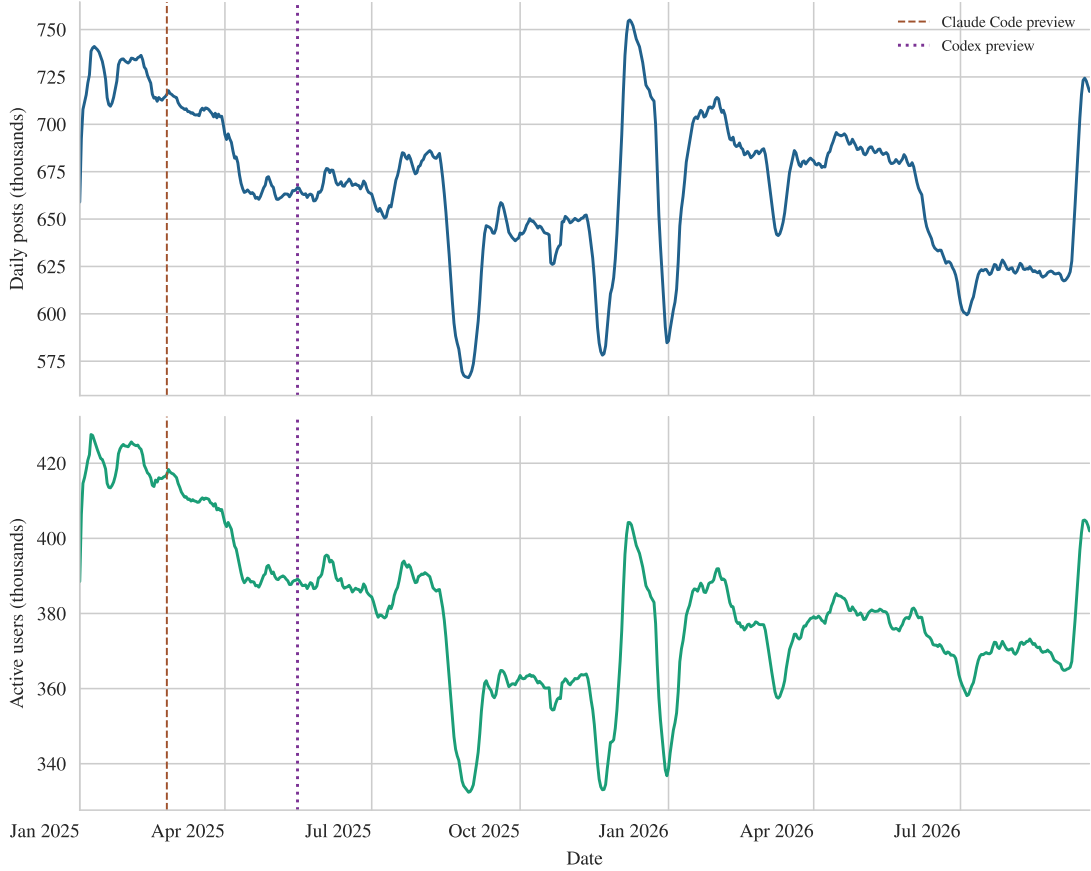


Figure 2: Observed daily posts and active accounts from January 2025 through September 19, 2026. Lines are seven-day trailing means; vertical markers show public research-preview announcements.

The median identifier contributed 2 posts in the full file, while the 99th percentile contributed 107. The most active 47,042 identifiers, representing 0.1% of identifiers, contributed 22.5% of observed posts. We found 1,213,143 identifier-days with at least 20 posts, and the highest daily count was 8,009. These values establish concentration, but high counts can reflect several human and automated practices.

Table 1 reports the immediate modeled changes under the planned 30-day linear specification. Both intervals include zero, and neither Holm-adjusted test rejects a zero immediate change at the 5% level. Removing identifier-days with at least 20 posts excludes 12.8% and 13.1% of posts in the two event windows; the resulting estimates are 0.14% and -0.47%. This check measures dependence on prolific days, not bot prevalence.

Table 1: Modeled immediate changes in observed daily posts around two research-preview announcements.

Announcement	Change (%)	95% interval (%)	Holm p
Claude Code, February 24, 2025	0.16	[-0.69, 1.01]	0.718
Codex, May 16, 2025	-0.62	[-1.64, 0.41]	0.473

The estimates depend on the window. Table 2 shows that the Codex estimate changes sign as the window expands. The 30-day quadratic specification gives 1.14% [-0.19%, 2.49%] for Claude Code and -1.59%

$[-2.97\%, -0.18\%]$ for Codex. These variants are exploratory; their intervals are not adjusted for the additional comparisons. The broader Codex windows include a decline in both posts and active identifiers during March and April 2025, followed by a flatter period. Without collection records, we cannot separate a change in observed coverage from a change in behavior. The planned 30-day result is consequently a description conditional on its trend and window, not a stable estimate of a product effect.

Table 2: Exploratory window sensitivity of modeled immediate changes. Each cell gives the percentage estimate and 95% interval under a separate linear model.

Days on each side	Claude Code (%)	Codex (%)
14	-0.05 [-0.39, +0.29]	-1.37 [-2.27, -0.46]
21	+0.71 [-0.28, +1.72]	-1.20 [-2.22, -0.17]
30	+0.16 [-0.69, +1.01]	-0.62 [-1.64, +0.41]
45	+0.23 [-1.58, +2.08]	+0.88 [-0.83, +2.63]
60	-0.61 [-3.07, +1.90]	+2.48 [+0.25, +4.76]

Figure 3 shows the daily observations and weekday-adjusted fitted trends from the two planned 30-day models. The omitted announcement date is at zero; fitted lines are shown separately on either side. Daily variation and changing background slopes limit what can be read from the estimated jumps. The numerical intervals in Table 1 describe uncertainty under the specified model.

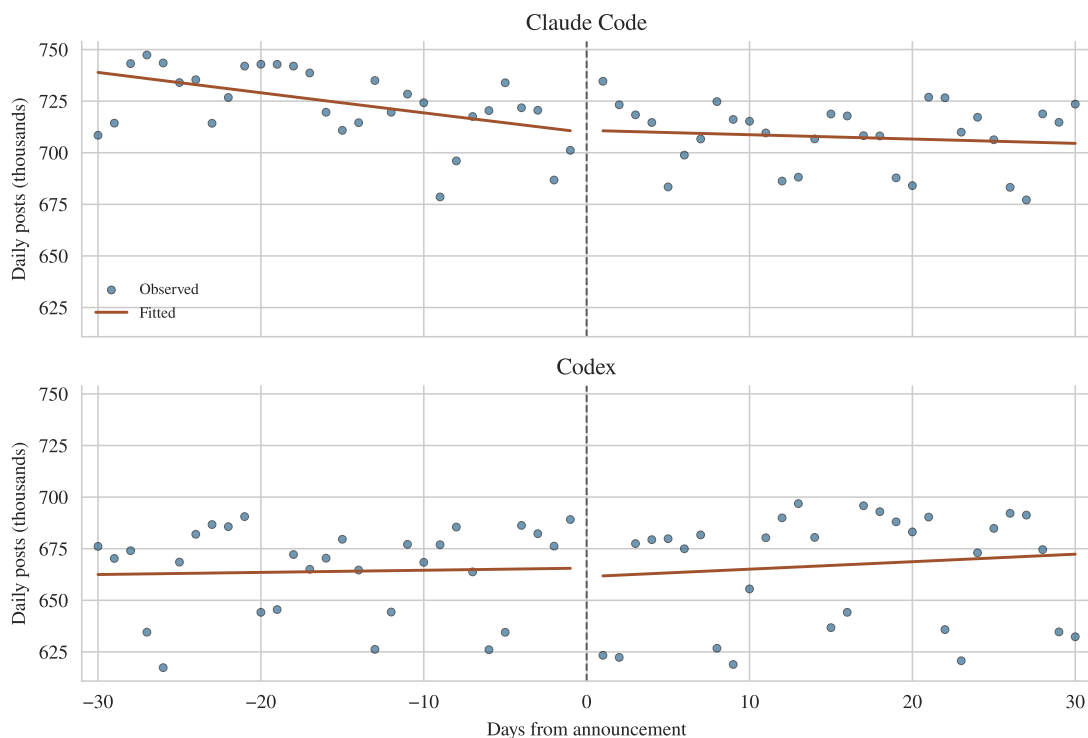


Figure 3: Observed daily posts and weekday-adjusted fitted trends in the 30 days on each side of the two announcements. The announcement day is omitted from each model.

4 Interpretation and next data needs

The observed volume and account concentration provide a baseline for studying posting behavior, but they do not identify bot activity. Automated accounts can post infrequently, and prolific accounts can be human. Publication of coding tools is also an indirect exposure measure: the file records no tool use, and these products do not uniquely govern the ability to automate Reddit posting. Other platform changes could occur in the same windows.

A stronger analysis would require documented collection coverage, post or comment type, community identifiers, timestamps within a day, content or validated bot labels, and a comparison series exposed to different adoption patterns. Those fields would permit tests of text generation, temporal regularity, and group-level heterogeneity. Until they are available, the supported conclusion is limited to the observed posting series and its uncertainty around the selected dates.

References

- [1] Gordon Burtch, Dokyun Lee, and Zhichen Chen. 2024. “The consequences of generative AI for online knowledge communities.” *Scientific Reports* 14. doi:10.1038/s41598-024-61221-0.
- [2] Zhen Sun, Zongmin Zhang, Xinyue Shen, Ziyi Zhang, Yule Liu, Michael Backes, Yang Zhang, and Xinlei He. 2025. “Are We in the AI-Generated Text World Already? Quantifying and Monitoring AIGT on Social Media.” *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 22975–23005. doi:10.18653/v1/2025.acl-long.1120.
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